Joshua Elieson

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Skills

Languages: JavaScript, TypeScript, Java, Python, C/C++, C#, SQL

Technologies & Tools: AWS, Docker, Kubernetes, Linux, Bash, Terraform, React, Next.js, PostgreSQL, Redis, Git, CI/CD, REST APIs, Microservices

Clearance: U.S. Citizen with active U.S. Government Security Clearance (Secret)

Education

Columbia University — Aug 2026 – Dec 2027
Master of Science in Computer Science — *Expected Graduation: Dec 2027*

University of Utah — Aug 2022 – May 2026
Bachelor of Science in Computer Engineering — *GPA: 3.84/4.0*

Work Experience

Northrop Grumman — Aug 2024 – Aug 2025
Software Engineering Co-op (Cloud Services and Integration) — *Roy, UT*

- Developed 10+ automated validation scripts in Jython and Groovy to scan thousands of UML system models for security and deployment errors. Reduced manual review time by 60% and prevented critical integration failures before release.
- Refactored and optimized a complex CI/CD pipeline, cutting deployment time from 8+ hours to under 10 minutes. Supported faster delivery cycles for 200+ internal projects.
- Built and maintained microservice components using enterprise patterns common in AWS and Java-based environments, improving system reliability and supporting seamless deployment across distributed teams.
- *AWS, Java, Jython, Groovy, C++, Git, CI/CD, Agile*

Northrop Grumman — May 2024 – Aug 2024
Software Engineering Intern (Physical Security) — *Roy, UT*

- Implemented back-end linear algebra algorithms in C to convert live incoming raw LIDAR data packets into structured point cloud (PCD) files, improving processing speed by 3x and delivering a smoother visualization for security and other developers.
- Developed front-end features in React to render interactive 3D point clouds, enabling non-technical users from 5+ internal teams to quickly analyze sensor data directly in a web application.
- *TypeScript, React, JavaScript, C, Node.js, REST APIs, Git, Linux, Jira, Agile*

University of Utah — Aug 2023 – May 2024
Teaching Assistant (Physics for Scientists and Engineers) — *Salt Lake City, UT*

- Led bi-weekly lab sessions and office hours for over 100 students, providing individualized support and fostering deeper engagement with core physics concepts throughout the semester.
- Designed and delivered over 20 interactive workshops and demonstrations that simplified complex topics like mechanics and electromagnetism, resulting in a 10% increase in final course grades among students.

Projects

- **Forge AI IDE — Summer 2026:** Developed an AI-powered integrated development environment leveraging OpenAI APIs, multi-agent systems, agent orchestration, shared context management, model routing, and real-time code generation. Engineered a scalable interface supporting concurrent AI agents, file-aware agent interactions, customizable layouts, and intelligent software development assistance.
  Next.js, TypeScript, Rust, React, Tailwind CSS, OpenAI API
- **WikiSpeedrunning.com — Spring 2026:** Built a full-stack web platform for collaborative route optimization and competitive speedrunning of Wikipedia and other wiki articles. Implemented ranked matchmaking, Elo ratings, leaderboards, route tracking, and real-time multiplayer game state synchronization. Designed scalable backend services and optimized database queries to support persistent user profiles and competitive gameplay.
  Next.js, TypeScript, PostgreSQL, Prisma, Redis, Supabase
- **Mechanical Detection of Magnetic Resonance — Undergraduate Research (2025 – 2026):** Designed and implemented software in Python and MATLAB to process, log, and visualize voltage signals from a custom magnetic-sensing device, achieving a 10x improvement in detection sensitivity from the initial design and demonstrating potential for next generation magnetic sensors and spin-detection technologies.
  Python, MATLAB, Signal Processing